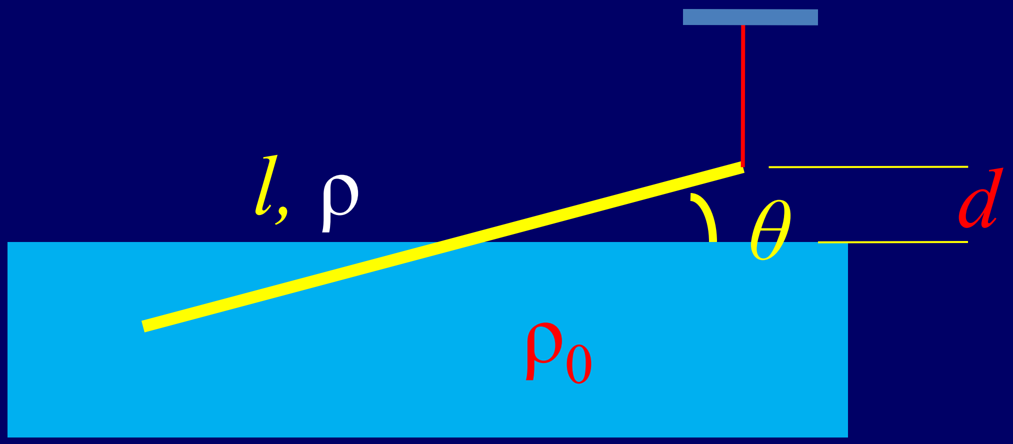
Quiz 10: A uniform rod with mass density ρ and length l is partly floating on the liquid with mass density $\rho_0(> \rho)$. The cross-section of the rod is S. When one end of the rod is hanged above the liquid surface with the height d . Calculate the angle between the rod and the liquid surface.

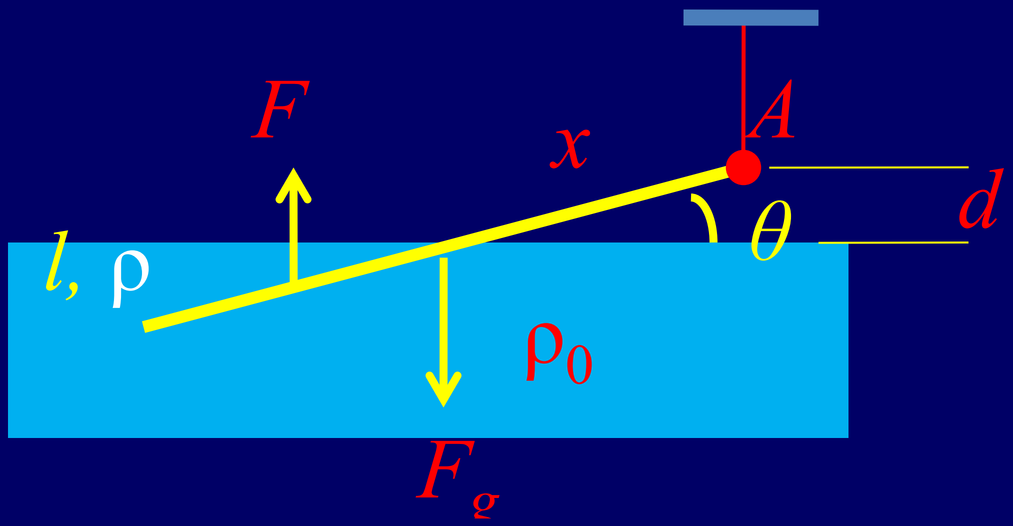


Solution:

Floating force: $F = \rho_0(l-x)Sg$

Gravitation force: $F_g = \rho l S g$

Due to equilibrium state, the torgue should be zero about Point A.



$$\tau_F = F\left(x + \frac{l-x}{2}\right)\cos\theta$$

$$\tau_g = F_g \frac{l}{2}\cos\theta$$

$$x = l\sqrt{\frac{\rho_0 - \rho}{\rho_0}}$$

$$\sin\theta = d / x$$